

ADR SEVERITY & CASE CLASSIFICATION REPORT

Prepared by: Dhruv Gupta | B.Pharm Student

1. Objective

To classify the five adverse drug reaction according to seriousness and expectedness, and summarize their clinical impact.

2. Classification Principles

Seriousness is based on outcomes such as death, life-threatening event, hospitalization, significant disability, congenital anomaly, or another medically important condition. Severity means intensity and is not the same as seriousness. Expectedness is judged against applicable reference safety information. The task image supplies expectedness labels for Cases 01–03 only; for Cases 04–05, expectedness must be verified rather than guessed.

3. Case Classification

Case / Drug	Reported reaction	Seriousness	Expectedness
01 – Amoxicillin	Skin rash, itching	Non-serious (given)	Expected (given)
02 – Ibuprofen	GI bleeding requiring hospitalization	Serious (given)	Expected (given)
03 – Metformin	Lactic acidosis	Serious (given)	Unexpected (given)
04 – Atorvastatin	Mild elevated liver enzymes	Non-serious (given)	Not specified; verify product information
05 – Levofloxacin	Tendon rupture	Serious (given)	Not specified; verify product information

4. Clinical Impact and Rationale

Case 01: Amoxicillin – Skin Rash and Itching

The task classifies this as non-serious and expected. Rash and itching can cause discomfort and may need clinical review. Monitor for progression, mucosal involvement, facial swelling, or breathing difficulty, which may indicate a more severe reaction.

Case 02: Ibuprofen – Gastrointestinal Bleeding

Classified as serious and expected in the task. Hospitalization is a seriousness criterion. GI bleeding can cause blood loss, anemia, circulatory instability, and may become life-threatening depending on extent.

Case 03: Metformin – Lactic Acidosis

Classified as serious and unexpected in the task. Lactic acidosis is a potentially life-threatening metabolic emergency requiring urgent assessment and treatment. The brief scenario does not establish patient-specific causality or risk factors.

Case 04: Atorvastatin – Mild Elevation of Liver Enzymes

Classified as non-serious in the task. Mild enzyme elevation without a serious outcome is treated as non-serious in this scenario. Expectedness is not shown and should be checked against the applicable product information.

Case 05: Levofloxacin – Tendon Rupture

Classified as serious in the task. Tendon rupture may cause major functional limitation and can require urgent assessment, immobilization, or surgery. Expectedness is not shown in the image and should be verified from product information.

5. Summary

The task labels three cases as serious (ibuprofen-associated GI bleeding, metformin-associated lactic acidosis, and levofloxacin-associated tendon rupture) and two as non-serious (amoxicillin-associated rash/itching and mild atorvastatin-associated liver enzyme elevation). Expectedness is explicitly given as expected for Case 01 and Case 02, and unexpected for Case 03. It is not specified for Cases 04 and 05.

6. Conclusion

ADR classification requires separate assessment of seriousness, symptom intensity, expectedness, and clinical impact. Hospitalization and potentially life-threatening outcomes are key seriousness indicators. Missing expectedness information should be verified from reference safety information rather than assumed.

7. References

1. VirtualWorks Internship, Task 4: Severity & Case Classification (case details and labels shown in the task dashboard).
2. World Health Organization. Pharmacovigilance. <https://www.who.int/teams/regulation-prequalification/regulation-and-safety/pharmacovigilance>
3. ICH E2A: Clinical Safety Data Management—Definitions and Standards for Expedited Reporting.